

Scaling Exercise: Corrections

1 Methods

I sweep over a large variety of hyperparameter configurations, including warmup, learning rate schedule, learning rate, batch size, and load balancing loss. My initial experimentation aimed to establish the general minima for the isoFLOPs curves for each FLOPs budget. Then, I investigated the effects of learning rate on the stability of the learning curve and the stability of the isoFLOPs curves. Following that, I moved to analyzing the correlation between final validation loss and isoFLOPs curve stability. Finally, I conduct a large experiment to identify the set of hyperparameters that induce the cleanest isoFLOPs curves.

2 Discussion

I make the following observations and analyses

- My baseline learning curves had some early training instability, which I hypothesized could be addressed by a warmup schedule.
- My baseline isoFLOPs experiments were producing points that were much noisier at the minimum, which prompted me to investigate the learning rate
- I found that I could tolerate a much higher learning rate (up to 7.5×10^{-4}) than my initial experiments (1×10^{-6}). Higher learning rates result in significantly lower validation loss by the end; however, I also observe that higher learning rates tended to correlate with noisier isoFLOPs curves, particularly near the minima. Thus, I conclude that **hyperparameters that produce lower validation loss do not necessarily correlate with cleaner isoFLOPs curves** (for this task)
- Upon careful review, I identified some correlating behavior between the average final validation loss of points near the minima in an isoFLOPs curve and the noise level of that isoFLOPs curve.

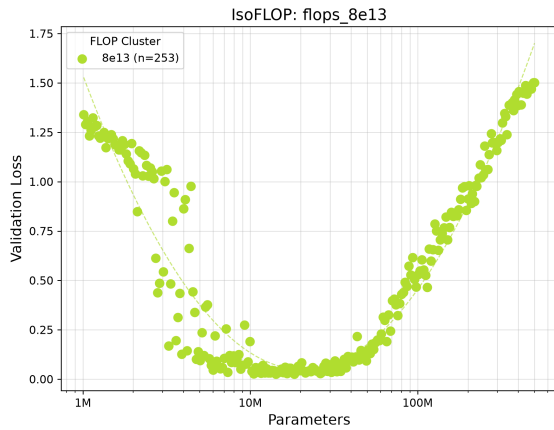


Figure 1: Instability around the 1.0 validation loss location ($\text{lr} = 1 \times 10^{-5}$)

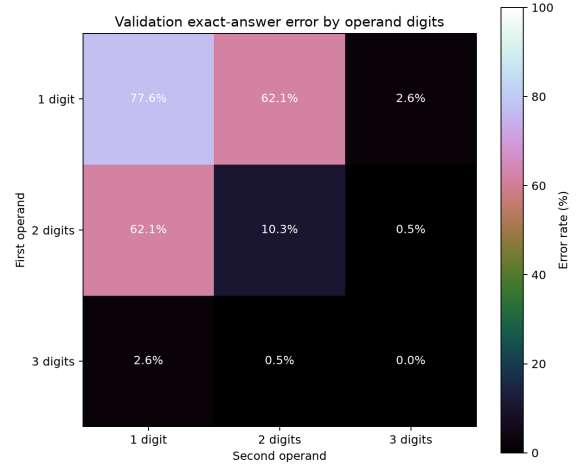


Figure 2: Heatmap of validation loss for saturated model

- I investigated this phenomenon by holding the hyperparameters constant and plotting curves at various FLOPs budget. I found 2 locations with high instability: the 1.0 validation loss and near ~ 0.1 validation loss.
 - I hypothesized this was a task specific result: there are 1M total unique datapoints for the 3-digit addition task. Additions involving lower budgets are much sparser in data: for example, there are only 100 are single digit addition data points. Thus, the saturation and unstable scaling at the 0.1 validation loss indicates the model is now trying to learn addition for 1 digit or 2 digit addition, which inherently happens slower since the data is much sparser.
 - I hypothesize the instability at lower model parameter sizes around the 1.0 validation loss is caused by smaller models hitting the 3-digit addition 'wall' that they need to train past to make progress. Models that make it past the wall successfully learn 3-digit addition and obtain drastically lower losses.
 - I verified my intuition by analyzing the distribution of validation errors of models that were trained to 0.1 and below validation loss w.r.t. the kind of addition (3-digit, 2-digit, etc.)
- Armed with this, I carefully selected FLOPs budgets with minima away from unstable locations and conducted a large hyperparameter search for the most stable isoFLOPs curves. I measured a fit's cleanliness by it's RMSE. My best configuration for both dense and MoE (4 experts with top-2 routing) is:

Hyperparameter	Value
Learning rate	2×10^{-5}
Batch size	64
Schedule	constant
Warmup fraction	0.05
Initial LR	0
Gradient clipping	0.8
Adam	β_1 0.9, β_2 0.95, ϵ 1×10^{-8}
MoE load balancing loss	4×10^{-3}
Dtype	bf16

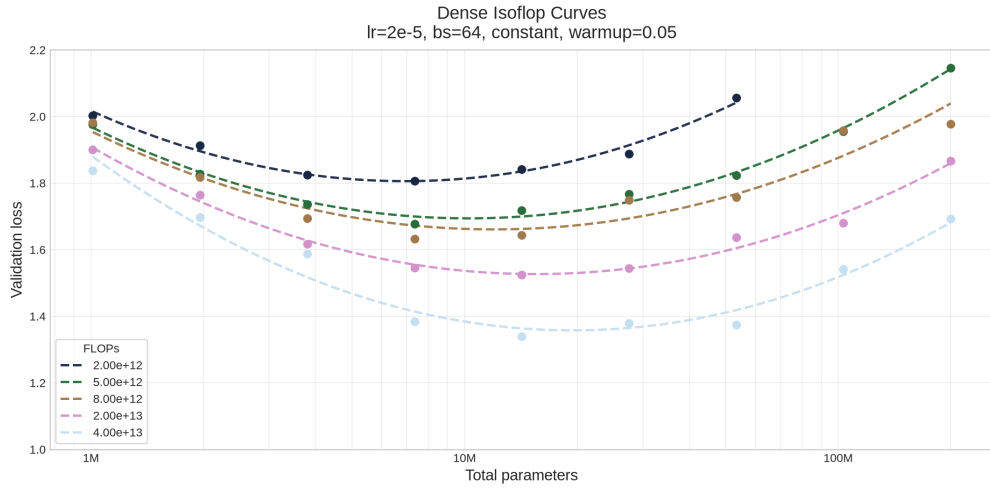


Figure 3: Dense isoFLOPs curves

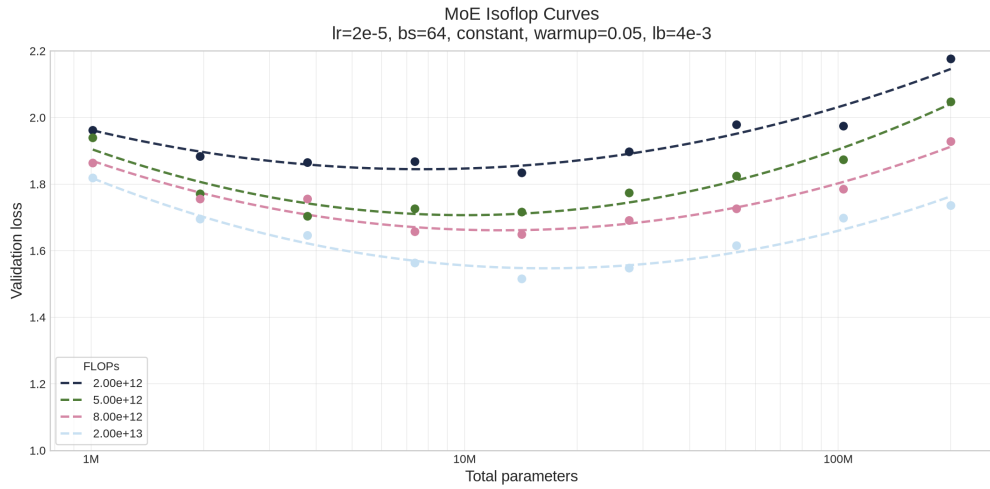


Figure 4: MoE isoFLOPs curves

The curves continue to have some noise but are cleaner than before.

3 Parametric Fits

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$$

Architecture / parameter basis	E	A	α	B	β	a	b
Dense	0.9613	880.4698	0.5096	11.0381	0.2748	0.3503	0.6497
MoE, total parameters	0.5116	169.5864	0.3892	5.2227	0.1497	0.2778	0.7222
MoE, activated parameters	0.9167	90.1147	0.3610	11.4705	0.2729	0.4305	0.5695

The fit implies that MoEs do seem to have more data hunger but they do not seem to scale better on parameters. I investigated the parameter scaling further by deriving the parametric fit over a wide variety of hyperparameter configs and observe that this trend seems to generally hold across multiple configurations.

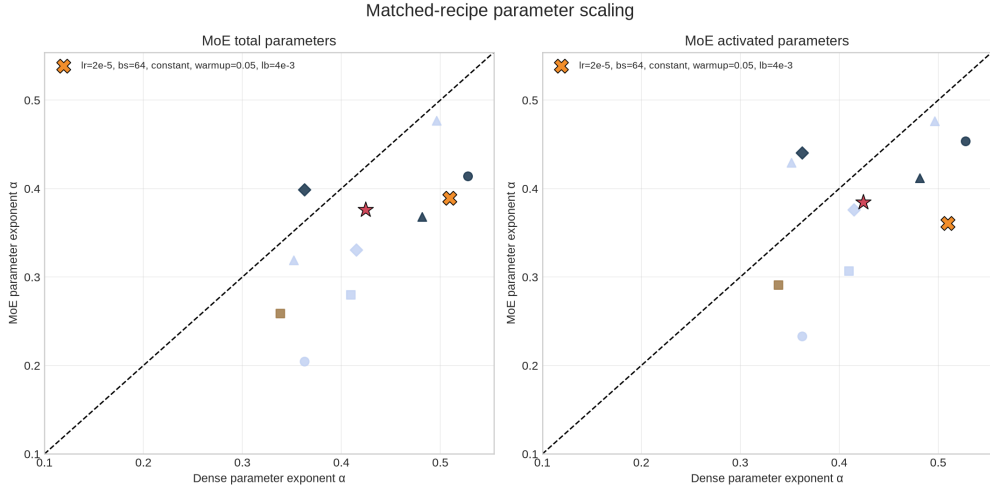


Figure 5: Comparison of α for MoE vs. dense across various hyperparameter configs